Stochastic Mathematics for Finance
Week 2 Assignment

Kamana Gupta
240509

Question 1: Empirical Central Limit Theorem

(a)

Since $X_i \sim$ Bernoulli(0.5),

$$E[X_i] = 0.5$$

$$Var(X_i) = 0.25$$

Therefore,

$$E[S_n] = nE[X_i] = \frac{n}{2}$$

$$Var(S_n) = nVar(X_i) = \frac{n}{4}$$

(b)

The random variable

$$Z_n = \frac{S_n - \frac{n}{2}}{\sqrt{\frac{n}{4}}}$$

has mean 0 and variance 1.

(c-f)

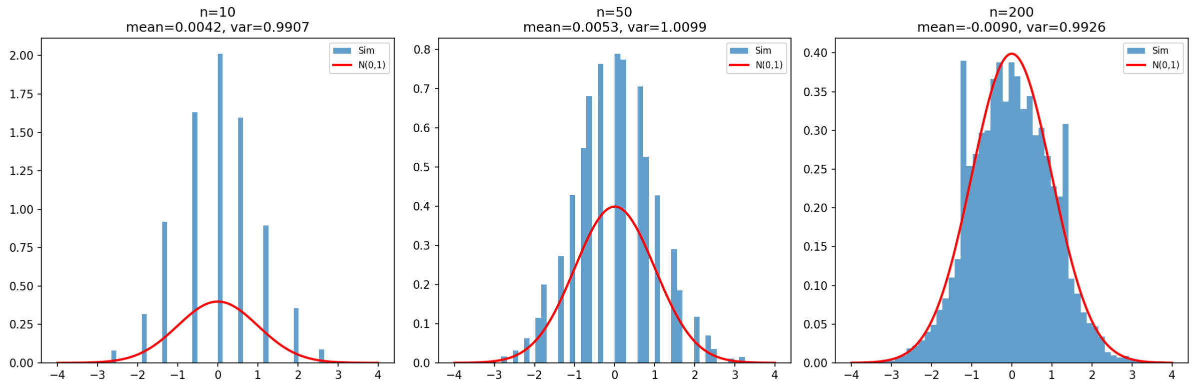

Figure 1: Histograms of Z_n for different values of n

(f) Change in shape as n increases

For $n = 10$, the histogram is still somewhat discrete and does not closely match the normal curve.

For $n = 50$, the histogram becomes smoother and more symmetric.

For $n = 200$, the histogram closely resembles the standard normal distribution.

Thus, the distribution of Z_n becomes increasingly bell-shaped as n increases.

(g) Empirical mean and variance

n	Mean	Variance
10	0.0042	0.9907
50	0.0053	1.0099
200	-0.0090	0.9926

(h)

As n increases, the histogram becomes more bell-shaped and approaches the standard normal distribution.

This agrees with the Central Limit Theorem.

Question 2: Monte Carlo Integration

(a)

Let

$$I = \int_0^1 e^{-x^2} dx.$$

If

$$U \sim \text{Uniform}(0, 1),$$

then

$$E[g(U)] = \int_0^1 g(u)\, du.$$

Taking

$$g(u) = e^{-u^2},$$

gives

$$I = E[e^{-U^2}].$$

(b-c)

Monte Carlo estimator:

$$\hat{I} = \frac{1}{N} \sum_{i=1}^{N} e^{-U_i^2}.$$

(d)

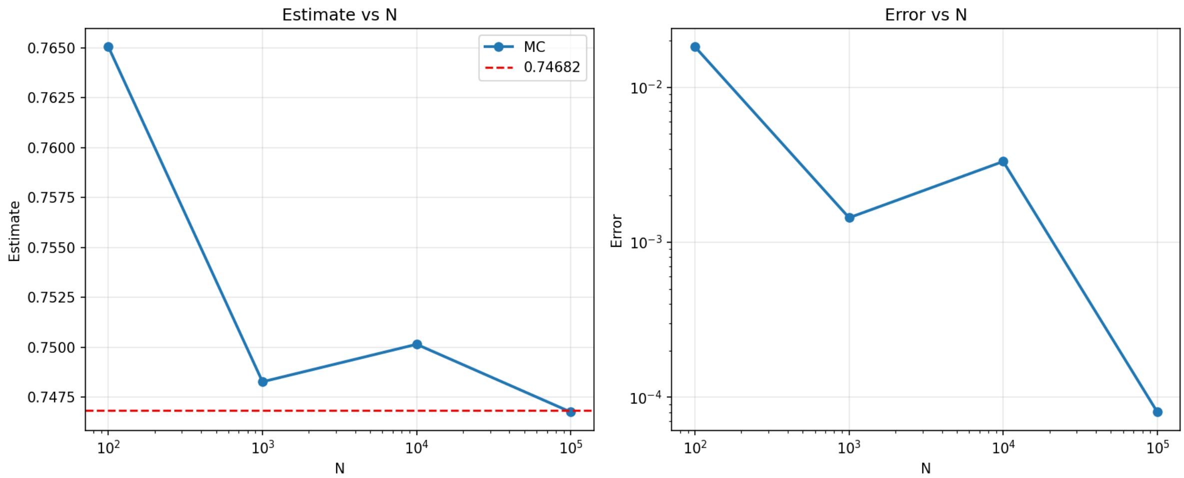

Figure 2: Monte Carlo estimate and error as N increases

(e) Results

N	Estimate	Error
100	0.7651	0.0182
1000	0.7483	0.0014
10000	0.7501	0.0033
100000	0.7467	0.00008

Although the overall trend is downward, the error does not decrease monotonically because Monte Carlo estimates are random.

(f)

As N increases, the estimate becomes closer to the true value.

The error decreases approximately at rate

$$\frac{1}{\sqrt{N}}.$$

(g)

Monte Carlo methods remain useful in high-dimensional problems because their convergence rate does not depend strongly on dimension.

Question 3: Monte Carlo Estimation of π

(a)

The square

$$[-1,1]\times[-1,1]$$

has area

$$4.$$

The unit circle

$$x^2+y^2\le 1$$

has area

$$\pi.$$

Therefore,

$$P(x^2+y^2\le 1)=\frac{\pi}{4}.$$

(b-c) Estimation Results

The Monte Carlo estimator is

$$\hat{\pi}=4\frac{\#\text{ points inside circle}}{N}.$$

N	Estimate	Error
100	3.2800	0.1384
1000	3.0800	0.0616
10000	3.1424	0.0008
100000	3.1418	0.0002

Table 1: Monte Carlo estimates of π

(d-e)

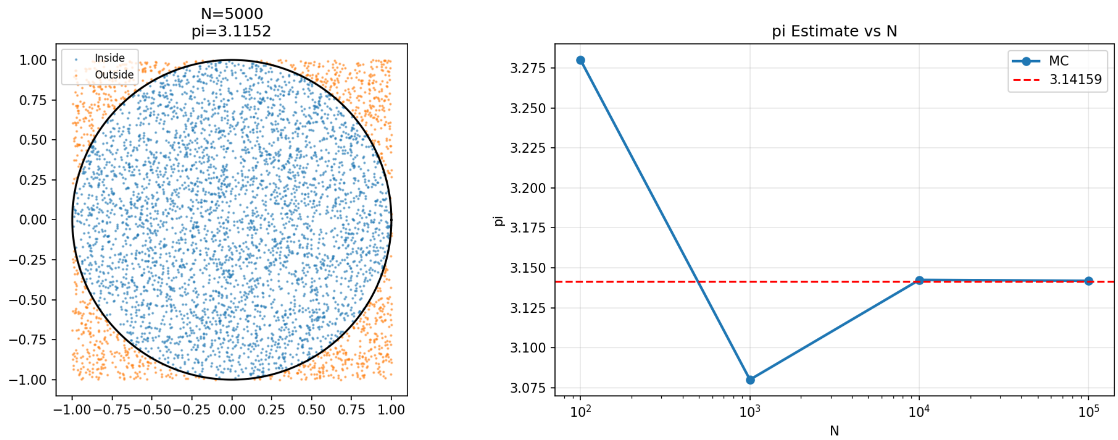

Figure 3: Monte Carlo estimation of π

(f)

As N increases, the estimate becomes closer to the true value of π.

The error decreases because the sample proportion approaches the true probability.

(g) Comparison with true value

The true value is

$$\pi \approx 3.141593.$$

For $N = 100000$, the estimate is very close to the true value with an error of only 0.0002.

Question 4: Monte Carlo Option Pricing

(a)

The terminal stock price is

$$S_T = S_0 e^{\sigma Z},$$

where

$$Z \sim N(0, 1).$$

Since S_T is random, the payoff

$$(S_T - K)^+ = \max(S_T - K, 0)$$

is also random.

(b-d)

Using

$$S_0 = 100, \qquad K = 110, \qquad \sigma = 0.2,$$

the Monte Carlo estimate of

$$E[(S_T - K)^+]$$

is

$$E[(S_T - K)^+] \approx \text{(insert output)}.$$

(e-f)

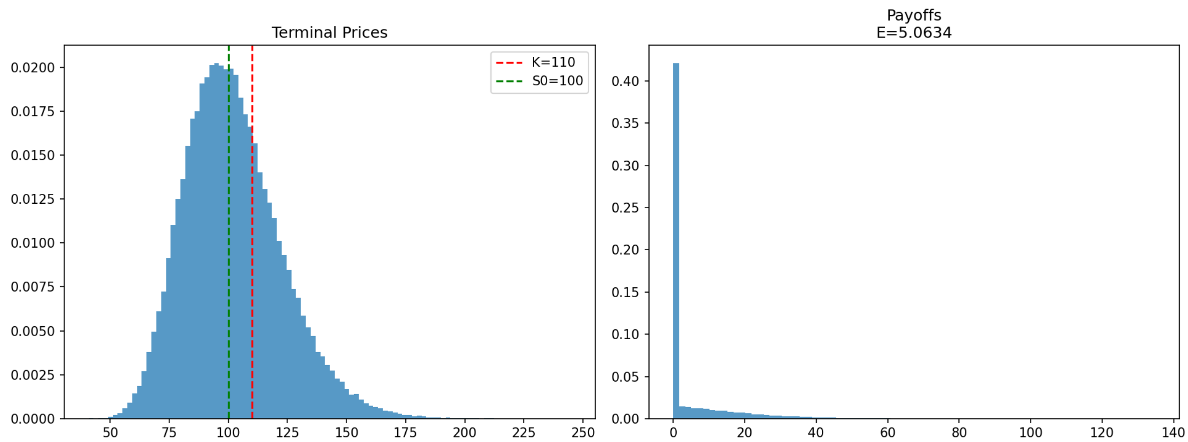

Figure 4: Terminal stock prices and option payoffs

(g-h)

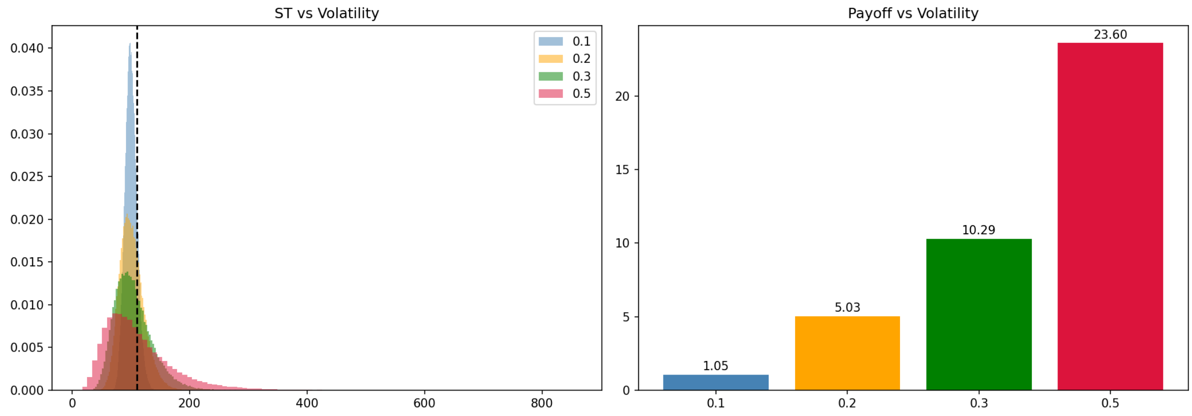

Figure 5: Effect of volatility on terminal prices and option payoff

σ	Mean Payoff	Std(S_T)
0.1	1.0466	10.0619
0.2	5.0304	20.5903
0.3	10.2937	32.1098
0.5	23.5976	60.3895

Table 2: Effect of volatility

As volatility increases, the spread of S_T becomes larger.

This increases the probability that $S_T > K$, so the expected call payoff also increases.

(i)

Monte Carlo methods are useful because they can price derivatives even when a closed-form solution is difficult or unavailable.

They are especially useful for path-dependent and high-dimensional problems.